

FREUID Challenge 2026 — Technical Report

1. Introduction

We address fraud detection in identity documents under a cross-domain private test regime (two undisclosed document types + recaptured images). Our guiding finding: **fine-tuned backbones overfit the seen countries and collapse on blind domains, while frozen pretrained backbones with lightweight LoRA adapters preserve the pretraining generalization prior.** The final system is a rank-mean ensemble of three frozen-backbone classifiers with an inference-side lever for recaptured images.

2. Method

2.1 Models

- **FB** — frozen DINOv2-L (`vit_large_patch14_dinov2.lvd142m`), LoRA on q/k/v of the last 8 blocks ($r=16$, $\alpha=32$), MLP head (hidden 256), input 448x714 (aspect resize), 5 folds. LoRA merged into plain weights after training.
- **FC** — frozen OpenCLIP ViT-L/336 (`vit_large_patch14_clip_336.openai_ft_in12k_in1k`), identical recipe, CLIP normalization. Decorrelated pretraining prior (image-text contrastive vs. self-supervised) is the source of the ensemble gain.
- **FD** — frozen SigLIP2-L/16-384 (`vit_large_patch16_siglip_384.v2_webli`), identical recipe, input 448x720, 0.5/0.5 normalization. A third, independently-pretrained prior (sigmoid image-text, WebLI).

All backbones are trained as 5-fold CV. The final pick uses all **5 FB folds** with hflip-TTA as its core; **FC3** (3 folds) and **FD3** (3 folds) score the captured subset only, so the captured ens3 reorder stays within the 6h/A100 inference cap (Sec. 4).

Blind leave-country-out (LTO) evidence for the frozen-vs-finetuned choice: fine-tuned FA collapses to 0.18 on held-out GUINEA; frozen FB reaches 0.030. Adding FC improves blind rank-mean by -35% (GUINEA) / -27% (EGYPT) and pooled OOF by -70%. Adding FD improves blind rank-mean further (-4.6% GUINEA / -54% EGYPT vs. the FB+FC pair; FD is the strongest single backbone on held-out EGYPT) and pooled OOF by another -57%.

2.2 Training

- Data: official FREUID training set ONLY (no external datasets for any submitted weight; pretrained backbones are Apache-2.0, see Sec. 7).
- 5-fold CV. Loss: BCE. AMP. Epoch budget 4 with early best-selection (frozen-backbone LoRA converges at epoch 0-1).
- Augmentation: standard photometric/geometric domain aug + tamper-synth ($p=0.3$) + G1 recapture-style aug. Heavier domain-randomization

(FACT amplitude-spectrum mix, Shades-of-Gray color constancy)
DESTROYED the fraud signal in ablation (train loss stuck at random plateau) and was rejected.

2.3 Inference pipeline

1. Per-backbone fold plain mean -> per-backbone score.
2. **Cross-backbone equal rank-mean over FB/FC/FD** (score-mean dilutes the weaker-calibrated backbone on out-of-domain data).
3. **Recaptured-row detection** by resolution: a resolution is native only if it is high-resolution (width ≥ 1000 px) AND either frequent ($\geq 0.5\%$ of rows) or a known native cluster; everything else — including frequent but low-resolution formats such as 840x530 — is captured. In the training data digital captures are always ≥ 1000 px wide while recaptured/downscaled images are smaller, so width cleanly separates the two acquisition types (a frequent 840x530 recapture format would otherwise be misread as native).
4. **capShift**: captured rows are shifted by $\Delta = 0.75$ in logit space (per backbone) — recaptured bona fide images otherwise inflate the fraud tail (APCER at 1% BPCER dominates the metric). Public-LB verified: 0.01870 -> 0.01524.
5. **Captured-internal reorder (ens3)**: within captured rows, order by the equal mean-rank of the three backbones (FB5 + FC3 + FD3). FC3/FD3 are scored on the captured subset only, in a TTA / noTTA / off tier chosen by the remaining inference budget at the observed captured fraction (captured ordering is robust to the TTA reduction, spearman 0.9958); capShift is CPU-side and always applies. The reorder mechanic is public-LB verified (0.01524 -> 0.01304 -> 0.01252 as the captured-reorder ensemble grew); the shipped ens3 orders captured rows by the three released backbones only (FB5 + FC3 + FD3). The intermediate 0.01304 / 0.01252 points additionally used recapture-specialist checkpoints that are not part of this release; the released package reproduces the ens3 composition, which keeps the captured lever within the inference budget.
6. Output = strict total order -> $(\text{pos}+0.5)/n$ rank values (metric is rank-only; ties score as constants).

The single final pick (**fb5**) runs this pipeline with an FB5 hflip-TTA core: step 1 aggregates the 5 FB folds, step 2 degenerates to the FB score order, and FC3/FD3 enter only through the captured reorder in step 5.

Precision: bf16 (rank-safety spearman vs fp32: FB 0.99992, FC 0.999799, FD 0.999627). Ensemble-level / cross-backbone TTA on the core was rejected by measurement — blind leave-country-out showed degradation (+27.8% on held-out EGYPT; hflip severely harms the CLIP backbone on documents). The FB5 core keeps hflip-TTA (benign for the DINOv2 backbone; part of the pick's frozen public composition), while FC3/FD3 touch only the captured subset.

3. Data

- Training: FREUID official training set (69,352 images), 5-fold splits. **No external dataset was used to train any submitted weight.**
- External public datasets were used for offline evaluation only (regression-guard battery; never seen by any training pipeline):

FantasyID, IDNet (eval holdout), MIDV-Holo, DLC-2021. Full list + licenses: Sec. 7.

- External model weights: Sec. 7.
- Validation protocol: blind leave-country-out (GUINEA fold, EGYPT fold) — the only offline signal that correlated with cross-domain behavior; pooled OOF does NOT predict LB (measured: best-OOF model was worst on LB).

4. Inference / Budget

- Evaluator (organizer-confirmed): 1x NVIDIA A100 40 GB, 24 CPU cores, ≤ 6 h on the hidden test set.
- End-to-end offline (`--network none`) measurement on a single A10G (a ~ 2 - $2.7\times$ slower proxy for the A100): 7,821 images in ~ 33 min for the full pipeline (~ 0.25 s/img). Projected to the full 142,818-image test set: ~ 10 h on A10G $\rightarrow$ **~ 5 h on A100 at a conservative /2.0 factor** (~ 3.5 h at a realistic /2.7), within the 6 h budget. Peak GPU memory < 23 GB (fits A100 40 GB).
- `torch.compile` is applied to the inference model (rank-preserving; raw scores shift $\sim 1e-2$ but the pipeline is rank-based end-to-end). It amortizes over the many batches of the full test set ($\sim +17\%$ throughput measured on A10G).
- The FC3/FD3 captured pass (25.8 / 27.7 ms/img/ckpt on A10G) touches captured rows only, in the budget tier of Sec. 2.3 step 5 (TTA up to $c \leq 9.8\%$, noTTA up to $c \leq 19.6\%$, off beyond — `capShift` still applies). Core cost (FB5 hflip-TTA) dominates and scales linearly with the image count, so the run stays within the cap regardless of the private captured fraction.

5. Results

Stage	Public LB
FB5 (frozen DINOv2-L + LoRA)	0.01870
+ <code>capShift</code> (delta 0.75)	0.01524
+ captured reorder, early ensemble ⁽¹⁾	0.01304
+ captured reorder, all backbones ⁽¹⁾	0.01252
+ captured ens3 reorder (released FB5+FC3+FD3, size-based routing)	0.01238
Container-native output (<code>docker run</code> , rank values, reproduces the above)	0.01245
Private fb5 pick (FB5 hflip-TTA + captured lever)	hidden until 2026-07-24

⁽¹⁾ These two intermediate public-LB points additionally used recapture-specialist checkpoints that are not part of this release. They are shown only to document the reorder mechanic's progression; the released package reproduces the ens3 (FB5+FC3+FD3) composition.

Blind LTO (held-out country, freuid score, lower=better):

Model	GUINEA	EGYPT
FA (fine-tuned)	0.18	—
FB (frozen)	0.0297	0.0647
FC (frozen)	0.0325	0.0656
FD (frozen)	0.0813	0.0143
FB+FC rank-mean	0.0194	0.0474
FB+FC+FD rank-mean	0.0185	0.0218

FD's country profile is the mirror of FB/FC (weakest on GUINEA, 4.5x strongest on EGYPT) — direct evidence of prior decorrelation; the 3-way rank-mean improves both held-out countries.

The public leaderboard rows above keep the FB-based ordering: the public main block is in-domain, where cross-backbone rank-mean measurably loses to the single-backbone/score-mean orderings (consistent with pooled OOF); the 3-backbone rank-mean targets the cross-domain private blocks, per the blind LTO table.

6. Reproducibility

- Repo: <https://github.com/hyun910219/freuid-challenge-2026>, commit SHA: 88c6686e899fbc07d0117b4be37b48b1fb00b520. Model weights are frozen throughout; the post-freeze commits are rank-preserving only (157ae380... = size-based captured routing; 88c6686... = container flag-independence: file_system DataLoader IPC + device auto-detect). No weight/architecture/training change at any point.
- Docker: no-network container, weights baked in; flat `/data in, / submissions/submission.csv out`. See README. Verified end-to-end at 157ae380 (`git clone -> docker build -> docker run --network none` on 7,821 public images -> valid, contract-compliant `submission.csv` offline); the later robustness commit does not alter the container contract or the output ranking.
- Training: one config per member, `src/train.py --config configs/<m>.yaml`.
- Deterministic postprocessing; GPU/compile nondeterminism shifts raw scores at $\sim 1e-2$ but the pipeline is rank-based (the offline container reproduces our submitted ranking at Spearman ≥ 0.9999).

7. External Resources

Resource	License	Use
DINOv2 ViT-L <code>lv142m</code> (timm/Meta)	Apache-2.0 (repo LICENSE verified)	FB backbone
OpenCLIP ViT-L/336 <code>openai_ft_in12k_in1k</code> (timm)	Apache-2.0 (OpenAI CLIP origin: MIT)	FC backbone
SigLIP2 ViT-L/16-384 <code>v2_webli</code> (timm/big_vision)	Apache-2.0 (big_vision repo LICENSE verified)	FD backbone
PyTorch 2.12 / timm 1.0.27 / albumentations 2.0.8 etc.	BSD/Apache/MIT	framework

Resource	License	Use
FantasyID (Idiap, zenodo 17063366)	CC-BY 4.0 + CC0	eval-only (offline battery)
IDNet (eval holdout subset)	CC-BY 4.0	eval-only (offline battery)
MIDV-Holo	CC-BY-SA 2.5	eval-only (offline battery)
DLC-2021 (zenodo 6466768 / 6792396)	CC-BY-SA 2.5	eval-only (offline battery)

No external dataset was used to train any submitted weight; the four datasets above were used exclusively as an offline evaluation battery and are cited per the competition's external-resource rules (Rules Sec. 6).